

A local glyph process suffices: an information-theoretic mechanism comparison for the Voynich manuscript

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Abstract

We ask which classes of text-generating mechanism are compatible with the measurable structure of the Voynich manuscript text. Using the Zandbergen–Landini transliteration (32,428 paragraph-text tokens on 207 pages) and a battery of information-theoretic tests with marginal-preserving null models, we establish four properties. (1) Ordering information between adjacent tokens sits at the token boundary: the last glyph of a token and the first glyph of the next share 0.146 bit of excess mutual information, whereas whole tokens share only 0.076 bit; in Latin, German and Hebrew prose cut to the same line shapes this hierarchy is reversed. (2) The text restarts at every line: the pair formed by the last token of a line and the first token of the next carries 0.005 bit, against 0.062 bit for adjacent pairs at the same sample size; in the prose controls the two are equal. (3) Token boundaries are 73.7% predictable from one glyph on each side (prose: 38.7% to 44.9%). (4) There is no measurable lexicon: tokens sampled from a glyph trigram model are attested in the corpus as often as held-out real tokens (0.76 versus 0.79), whereas every language and every chunked language retains a large gap. A glyph automaton of order 3–4 with parameters that drift by section or page, restarted at each line and with the space as a symbol, reproduces every statistic of the battery, generalizes to held-out folios, and leaves a compression residual of -0.039 bit per glyph, within its own noise. Table-and-grille and copy-and-modify generators fail the boundary coupling by more than an order of magnitude; verbose and ciphertext-feedback substitutions of Latin fail the lexicon and residual tests; the Naibbe cipher matches the entropy profile and the residual but fails boundary coupling, hapax richness and page locality, and state-dependent variants buy boundary coupling only at the price of exposing plaintext structure. Page statistics drift smoothly with folio distance and follow the scribal hand more than the illustration section; labels behave as unique identifiers; and within a single hand and section the text carries no measurable information about the colour composition of the illustrations, with a sensitivity of roughly ten image-determined tokens per page. All tested observable structure is compatible with a local, low-order glyph generator; neither lexical structure nor image-related semantics is measurable beyond it. The results constrain what a solution could be; they do not show that the text is meaningless.

Keywords: Voynich manuscript; mutual information; null models; verbose cipher; Naibbe cipher; text generation; Markov models; Mantel test.

1. Introduction

The Voynich manuscript (Beinecke MS 408) has been analysed statistically for five decades without an accepted reading. Currier (1976) observed that its text falls into at least two statistical “languages” that coincide with different hands and that the physical line behaves as a functional unit; D’Imperio (1978) collected the early evidence. Later work established a low conditional character entropy and a rigid word structure (Landini 2001; Reddy and Knight 2011; Lindemann and Bower 2020; Bower and Lindemann 2021), long-range co-occurrence structure interpreted as evidence for content (Montemurro and Zanette 2013), and palaeographic evidence for five scribes whose distribution matches the statistical languages (Davis 2020). Explanations fall into three classes: a natural language written in an unknown script or cipher; a constructed system; and a meaningless text produced by a mechanical or quasi-mechanical procedure, for which concrete generators have been proposed (Rugg 2004; Schinner 2007; Timm and Schinner 2020). The cipher class has recently received explicit support in the form of constructive counter-examples: Greshko (2025) describes the Naibbe cipher, a verbose homophonic substitution executable with fifteenth-century materials that maps Latin or Italian onto Voynich-like ciphertext while reproducing several

of its statistics, and Kinnison (2026) reports that some verbose ciphers reproduce the manuscript’s unusual positional entropy profile. At the same time, Rozanova and Temerev (2026) argue from mutual information that the units of Voynichese are not letters, words and spaces in the usual sense, and Parisel (2026) reports layered positional and directional constraints.

These results raise a question that is more tractable than decipherment: which generating mechanisms can reproduce *all* of the measurable properties of the text at once? Individual statistics such as entropy or word-length distributions are known to be imitable by several mechanisms. A discriminating test must therefore (a) use a common battery of measurements applied identically to the manuscript and to candidate generators, (b) control for the confounds that are specific to this corpus (line structure, page-level drift, scribal hand, uncertain transcription of spaces), and (c) include a positive null model, that is, a mechanism that actually reproduces the whole battery, so that residual structure can be quantified rather than asserted.

This paper contributes such a battery and such a null model. Its main results are the following. First, the ordering information between adjacent tokens is concentrated in the glyphs at the token boundary rather than in token identities, and the text restarts at every line; both properties are the opposite of what prose controls show under identical cutting (Section 4.2–4.3). Second, token boundaries are predictable from the flanking glyphs to a degree matched only by spaces inserted into a glyph stream by rule (Section 4.4). Third, there is no measurable lexicon: tokens generated by a glyph trigram model are attested in the corpus as often as real held-out tokens (Section 4.5). Fourth, a glyph automaton of order three or four with parameters that drift by section or page, restarted at each line, reproduces every statistic in the battery, generalizes to held-out folios, and leaves no compression residual beyond its own sampling noise (Sections 4.6–4.8). Fifth, the published generators of meaningless text, several substitution ciphers of Latin and Italian including the Naibbe cipher, and a state-dependent extension of it each fail at least two statistics, and the pattern of failures is informative about what a cipher would need (Section 4.7). Sixth, page-level statistics drift smoothly with folio distance and follow the scribal hand more than the illustration section (Section 4.9), labels behave as unique identifiers (Section 4.10), and within a single hand and section the text carries no measurable information about the colour composition of the illustrations, with a quantified sensitivity (Section 4.11).

We are careful about what these results do and do not show. They establish that all tested observable structure is compatible with a local, low-order glyph process, and that neither lexical structure nor image-related semantics is measurable beyond it. They do not prove that the text is meaningless or non-linguistic: an encoding that pushes exactly this information below the sensitivity of our tests remains logically possible, and Section 5 spells out what such an encoding would have to do.

2. Materials

2.1 Transliterations and preprocessing

The primary source is the Zandbergen–Landini transliteration ZL3b in the IVTFF format with the extended EVA alphabet (Zandbergen and Landini 2025), 5,385 loci on 227 pages. All main analyses use the paragraph text (locus type P; 4,130 lines on 207 pages), excluding labels, circular and radial text. Inline markup is resolved as follows: alternative readings take the first alternative; high-ASCII codes and unreadable glyphs mark a token as uncertain, and tokens containing them are dropped; drawing intrusions and comments are removed; filler characters are deleted. IVTFF distinguishes certain spaces (“.”, 35,750 occurrences) from uncertain ones (“,”, 9,316 occurrences). In the baseline uncertain spaces are treated as no space; all key statistics are also reported with uncertain spaces treated as spaces (Table 8). After preprocessing the baseline corpus has 34,886 tokens and 6,875 types. Loci whose type is prefixed by “&” (labels placed in drawings, for example the zodiac nymph labels) are treated as labels; an earlier version of our parser had missed them, which

affected only the label analysis of Section 4.10. The Takahashi transliteration (IT2a, 34,384 tokens after the same preprocessing) is used as an independent check of transcription dependence.

Page metadata are taken from the ZL page headers: Currier language (\$L), illustration type (\$I: herbal, astronomical, zodiac, biological, cosmological, pharmaceutical, stars/recipes, text-only), quire (\$Q) and scribal hand (\$H, the palaeographic assignment of Davis 2020). Folio order is taken from the folio numbers; it is the binding order and not necessarily the order of production.

2.2 Reference corpora

Controls are classical Latin prose (Caesar, *De bello Gallico*, books 1–7, from The Latin Library), Italian verse (Dante, *Divina Commedia*), German verse drama (Goethe, *Faust I*), English prose (Tolstoy, *War and Peace*, used in Table 1 only) and consonantal Hebrew (Genesis, cantillation and vowel points removed), the latter as an abjad control with short tokens. Texts are lower-cased and reduced to letters. Unless stated otherwise, a control corpus is cut into “lines”, “paragraphs” and “pages” that follow the Voynich paragraph-text line-length, paragraph-length and page-length distributions exactly, so that every estimator sees the same sample structure. We note the obvious limitation that none of these is a medieval herbal or a heavily abbreviated Latin text (Section 6).

2.3 Illustration features

For the test of Section 4.11 we computed colour features of the illustrations directly from the Beinecke digital facsimile (folios 1r–58v, which are free of fold-outs; folio 12 is missing from the manuscript, and the page-to-folio assignment was verified visually on f26r, f27r and f58r). Each page image is rendered at 50 dpi, cropped by 6% on each side, and pixels are classified relative to the page’s median (parchment) colour into green, blue and red–brown classes. Features per page are the area shares of the three classes, their vertical and horizontal centroids, the mean hue of the green class, the blue and red shares of all coloured pixels, and the total coloured share (ten features, standardized). These are crude proxies for the content of the drawings (leaf mass, flowers, roots) and are meant to test one specific hypothesis, namely that the text of a page describes visible properties of the plant drawn on it.

3. Methods

3.1 Units and notation

A *glyph* is one EVA character; a *token* is a maximal string of glyphs between spaces as transcribed; a *line* is a transcribed line; *inner* tokens are those that are neither first nor last in their line. We use “token” rather than “word” because whether tokens are words is one of the questions at issue. Lines with fewer than two tokens are excluded from the battery (32,399 tokens remain). Entropies are in bits.

3.2 Mutual information with marginal-preserving null models

All mutual-information statistics are plug-in estimates of $I(X; Y)$ over a set of pairs, corrected by a null model. Plug-in estimates are biased upward for sparse joint distributions (Paninski 2003), and the bias depends on sample size and on the marginals; we therefore never interpret raw values, but only the *excess* of the observed value over the mean of null samples that preserve the marginals exactly. Two nulls are used. The *within-line permutation* null permutes the inner tokens of each line among themselves, which destroys order while preserving every line’s composition and every positional marginal; it is the null for the question whether order within a line carries information. The *pairing* null keeps the list of first elements fixed and permutes the list of second elements, which preserves both marginals of the pair set; it is used for pairs that do not live inside one line (line-break pairs, token–page pairs, pairs of consecutive lines). Comparisons between corpora are always made at equal numbers of pairs, by subsampling where necessary, because the residual bias of the

excess estimator still depends on n . A projection of the pair, for example (last glyph of the first token, first glyph of the second token), is a deterministic function of the pair, so its true MI cannot exceed the token-level MI; when a projected estimate exceeds the token-level estimate, the difference reflects the attenuation of the sparse token-level estimator, and the projected value is a lower bound on the true token-level MI.

3.3 Line breaks

For consecutive lines k and $k+1$ of the same paragraph we compare the excess MI of the pair (last token of k , first token of $k+1$) with that of adjacent within-line pairs subsampled to the same n (eight subsamples). We also compute, for consecutive lines of the same page, the excess MI of all combinations of position classes (first, middle, last), and the page specificity $I(\text{token}; \text{page})$ of tokens by position in the line.

3.4 Boundary predictability

Spaces are removed from each line and a model $P(\text{boundary} \mid k \text{ glyphs to the left}, k \text{ glyphs to the right})$ is fitted on a random half of the lines and evaluated on the other half. We report the information gain $1 - H(\text{boundary} \mid \text{context})/H(\text{boundary})$ and the F_1 score at threshold 0.5. As a control for what a rule-inserted space looks like, the Latin corpus is stripped of spaces and re-spaced by a fixed rule (a break after m , s , t or y once the current token has at least three letters).

3.5 Lexicon test

A within-token glyph model with two glyphs of context (a trigram model with token start and end symbols) is fitted on a random half of the tokens. Tokens are then sampled from the model. We compare the share of held-out real tokens (and types) whose string occurs in the training half with the same share for the sampled tokens. If the token inventory is an open set defined only by local glyph constraints, sampled tokens are attested as often as real ones; if there is a lexicon that restricts which strings occur, real tokens are attested far more often. We also report the mean log-probability per glyph of both sets under the model.

3.6 Entropy profile and residual redundancy

Treating each line as a string with the space as a symbol, we compute the plug-in conditional entropies $h(k) = H(x \mid k \text{ preceding symbols})$ for $k = 1$ to 5, and compare them with synthetic twins: texts sampled from a Markov model of order 1, 2 or 3 fitted to the same corpus, with the same number and lengths of lines. Plug-in $h(k)$ is biased downward for large k , so only the comparison with the twin, which shares the sample size, is interpretable. As a measure of redundancy at all ranges we use lzma compression (preset 9), in bits per symbol, and define the *residual* of a text as its lzma rate minus the lzma rate of an order-3 twin whose parameters are fitted separately for each combination of Currier language and illustration section of that text (a “section-drift” twin). A negative residual means the text contains redundancy that the drifting order-3 automaton does not capture. The residual of the automaton’s own samples (about ± 0.03 bit) sets the noise floor.

3.7 Candidate generators

All generators produce text in exactly the line-length, paragraph and page structure of the Voynich paragraph text, so that the battery sees identical sample structure. (i) *Glyph automata*: Markov models over glyphs including the space, of order 2 to 5, fitted per Currier language and section (“section drift”) and sampled line by line with restart at a start symbol; a *page-drift* variant interpolates page-level counts with section-level counts with weight λ (this variant sees the page’s own text and is in-sample by construction; it is used only to ask whether the residual page locality is explainable by page-level parameter variation). (ii) *Table-and-grille*, after Rugg (2004): tokens are prefix + midfix + suffix strings drawn from the empirical inventories obtained by a slot

decomposition of the attested tokens, arranged in a table of 40 columns and about 0.9 cells per token per section; a line is a run of consecutive cells in a row, the next line is the next row. (iii) *Copy-and-modify*, after Timm and Schinner (2020): each new token copies a source token (from the line above at a nearby column with probability 0.6, the previous token in the line with 0.2, any token of the page with 0.15, otherwise a seed token) and applies zero, one or two modifications from a list of glyph substitutions within similarity classes and prefix or suffix additions and removals, with a length-stabilizing rule. (iv) *Verbose cipher with fixed homophone choice*: each Latin letter maps to one to three glyph strings, the homophone being chosen deterministically by the previous glyph; spaces are re-inserted by a glyph rule; one independent Latin sentence per line. (v) *Ciphertext-feedback substitution*, $c_t = T[c_{t-1}, p_t]$ with random injective rows over 27 glyphs, in one- and two-glyph-per-letter forms, spaces by rule, independent sentences per line. (vi) *Naibbe cipher*: reimplemented from the published tables and the documented procedure (Greshko 2025): the plaintext is reduced to 23 letters ($w \rightarrow uu$, $j \rightarrow i$, $k \rightarrow c$), cut into tokens of one or two letters with unigram probability $17/36$, and each letter is enciphered by one of six tables (alpha, beta1–3, gamma1–2) selected by drawing from a shuffled 52-card deck (20/8/8/8/4/4); a one-letter token becomes a unigram string, a two-letter token the concatenation of a prefix string and a suffix string; 3% of ciphertext spaces are removed at random. Variants use one card per token, no space removal, running text instead of independent sentences, and Italian plaintext. (vii) *State-dependent Naibbe*: as (vi), but with probability w the table for the next token is drawn from a fixed random distribution conditioned on the last glyph of the previous token instead of from the deck. (viii) *Latin controls*: running text; rule-spaced text; and one independent sentence per line without any cipher. The Naibbe reimplementation was not checked against the author’s own outputs (Section 6).

3.8 Drift, hand and section

For pairs of pages of the same Currier language we compute the Jensen–Shannon divergence between their glyph-bigram distributions (450-glyph samples, additive smoothing 0.05) as a function of folio distance, and group pairs by whether they share the illustration section and the scribal hand. Within Herbal-B, the only section written by several hands, we compare same-hand and different-hand pairs at fixed folio distance.

3.9 Labels

For each illustrated section, the share of label tokens whose type occurs on at least one other page of the section is compared with the same share for equally many randomly drawn paragraph-text tokens of the same pages (or, where a page has too little text, of the same section), averaged over 20 draws.

3.10 External information: text versus illustration

For herbal pages of one hand, we compute the matrix of text distances between pages (Hellinger distance between relative token-frequency vectors, and separately between glyph-bigram frequency vectors) and the matrix of Euclidean distances between the standardized image-feature vectors of Section 2.3. Their Spearman correlation over page pairs is tested with a Mantel permutation test (Mantel 1967) that permutes the page identities of the image matrix; pairs of pages closer than three folios are excluded to remove local drift, and for page sets with several hands the permutation is stratified within hand. The sensitivity of the test is calibrated by adding to each page k pseudo-tokens that are deterministic functions of its image features (four binary features: blue share, green area, red share, vertical centroid), for $k = 2, 5, 10$ and 20 . As a secondary analysis we predict the illustration section of a page within one hand from its token frequencies and from its glyph-bigram frequencies (leave-one-page-out nearest centroid, excluding folio neighbours within a radius), against the majority rate and against prediction from the nearest remaining folio; this analysis is confounded by folio position because sections occupy contiguous folio ranges, and is reported with that caveat.

3.11 Robustness and uncertainty

Every core statistic is recomputed on the Takahashi transliteration, with uncertain spaces treated as spaces, on the held-out even-numbered pages with the automaton fitted to the odd-numbered pages (and the reverse), on the pages of each scribal hand with at least 30 pages, and on each Currier language. Standard errors of the core Voynich statistics are obtained by a leave-one-quire-out jackknife over the 16 quires (a page-level bootstrap with replacement is invalid for type-token statistics because duplicated pages create spurious repeats). All code, the parser, the generators and the logs from which every number in this paper is extracted are provided (Data and code availability).

4. Results

4.1 Basic statistics

Table 1 reproduces the well-known picture (computed with uncertain spaces treated as spaces, hence 34,886 tokens; the baseline treatment gives 32,399 tokens and 7,832 types, Table 8). At the token level the Voynich text resembles a natural language: 6,875 types, a hapax share of 0.67 and a Zipf slope of -0.78. At the glyph level it does not: the conditional bigram entropy is only 0.54 of the unigram entropy, against 0.77 to 0.85 in the controls, and this ratio cannot be brought to the range of the controls by any re-segmentation of the glyph stream. Applying identical byte-pair merges (Sennrich, Haddow and Birch 2016) to all corpora leaves the controls in the range 0.72 to 0.85 at every inventory size, whereas the Voynich ratio rises only from 0.54 to 0.74 after 80 merges, at an inventory of 105 units and two units per token (Figure 1, Table S4). This is a first indication that the glyph stream is not a re-encoding of letters.

Table 1. Basic token- and glyph-level statistics. Reference corpora are contiguous chunks matched to the Voynich token count (Hebrew: whole text). h_1 = unigram entropy, h_2 = conditional entropy given the previous symbol (bits per symbol, space included). Zipf slope: log-log slope over the 200 most frequent types.

Corpus	Tokens	Types	TTR	Hapax	Mean length	h_1	h_2	h_2/h_1	Zipf slope
Voynich, paragraph text	34,886	6,875	0.197	0.67	4.96	3.87	2.10	0.54	-0.78
Currier A	11,246	3,140	0.279	0.68	4.75	3.83	2.14	0.56	-0.79
Currier B	23,091	4,724	0.205	0.68	5.07	3.87	1.99	0.51	-0.81
Latin (Caesar)	34,886	8,426	0.242	0.59	6.10	4.02	3.30	0.82	-0.75
German (Faust)	30,959	6,221	0.201	0.63	4.78	4.13	3.19	0.77	-0.95
Italian (Dante)	34,886	6,574	0.188	0.62	3.92	4.01	3.15	0.79	-0.94
English (War and Peace)	34,886	4,416	0.127	0.50	4.43	4.12	3.33	0.81	-0.95
Hebrew (Genesis, consonantal)	17,789	6,319	0.355	0.65	4.41	4.13	3.53	0.85	-0.65

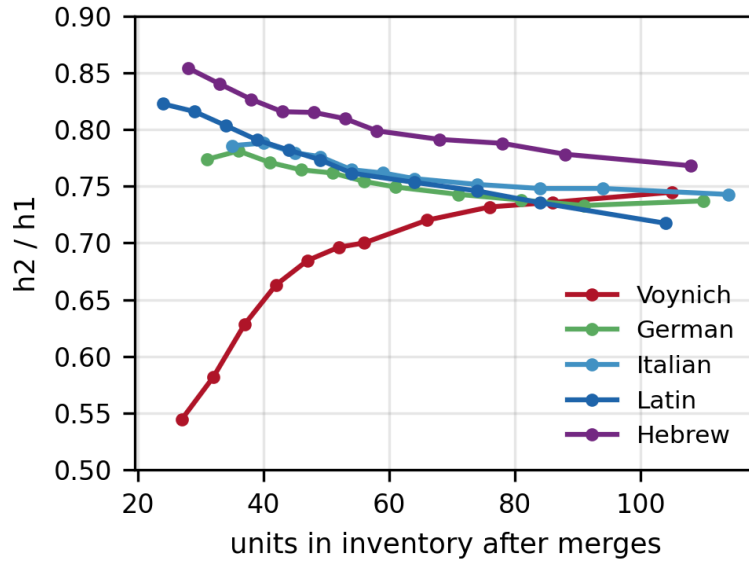


Figure 1. Ratio of conditional to unconditional entropy, h_2/h_1 , after identical byte-pair merges applied to the Voynich text and to four control corpora. The controls are invariant under re-segmentation; the Voynich text approaches them only when the inventory exceeds 100 units.

4.2 Ordering information sits at the token boundary

Table 2 and Figure 2 give the excess mutual information between adjacent inner tokens relative to the within-line permutation null, for the whole token pair and for projections of it. In the control corpora the token-level value is largest (0.242 bit in Latin, 0.216 in German, 0.178 in Hebrew, 0.409 in Italian) and the pair of boundary glyphs carries little (0.037, 0.056, 0.047; Italian 0.152 because of its clitics and elisions). In the Voynich text the hierarchy is inverted: tokens share 0.076 bit, but the last glyph of one token and the first glyph of the next share 0.146 bit, and the last two and first two glyphs 0.184 bit. Since the projection cannot exceed the true token-level MI, the token-level estimate is strongly attenuated by sparsity and the true value is at least 0.146; the point is not that the Voynich text has less ordering information than prose, but that its ordering information is carried by the glyphs at the boundary. Currier B shows the effect more strongly (0.187) than Currier A (0.082). Treating uncertain spaces as spaces raises both values (0.182 and 0.127) without changing the picture. These results agree with those of Rozanova and Temerev (2026), who report about 0.2 bit at the glyph boundary and a token-to-token predictability below one percent of token entropy.

Table 2. Excess mutual information (bits) between adjacent tokens inside lines, for different projections of the token pair, relative to a within-line permutation null with identical marginals. Line-initial and line-final tokens are excluded. Reference corpora are cut into the Voynich line-length distribution.

Corpus	token → token	last glyph → first glyph	last two → first two	last glyph → token	token → first glyph
Voynich (all)	0.076	0.146	0.184	0.138	0.142
Currier A	0.047	0.082	0.109	0.076	0.077
Currier B	0.081	0.187	0.227	0.159	0.172
all, uncertain spaces as spaces	0.127	0.182	0.235	0.174	0.191
Latin	0.242	0.037	0.212	0.234	0.167
Italian	0.409	0.152	0.422	0.337	0.356
German	0.216	0.056	0.173	0.191	0.156

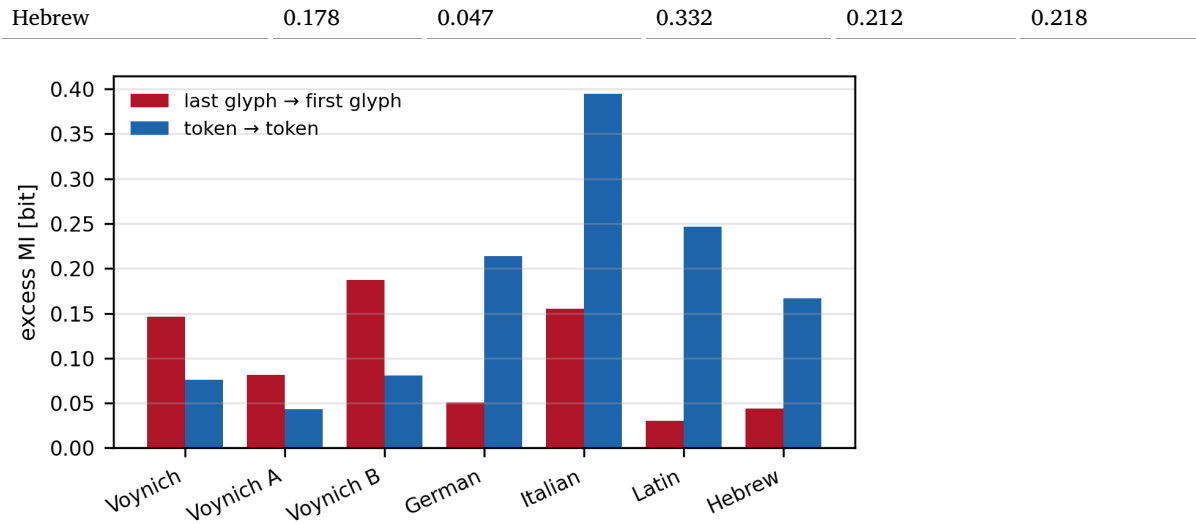


Figure 2. Excess mutual information between adjacent inner tokens (blue) and between the last glyph of a token and the first glyph of the next (red), relative to a within-line permutation null with identical marginals.

Two further observations characterize the boundary information. First, the tokens that follow each other more often than the null predicts are pairs like *chedy qokeey*, *shedy qokedy*, *ol chedy*: a token ending in *y* tends to be followed by one beginning with *qo*, a token ending in *l*, *r* or *s* by one beginning with *a*, *ch* or *sh*. Decomposing the excess MI by the edit distance between the two tokens under a global marginal-preserving null, the Latin excess of 0.262 bit lies almost entirely in dissimilar pairs (edit distance ≥ 3 : 0.243), that is, in collocations of different words, whereas the Voynich excess of 0.203 bit lies in pairs at edit distance ≤ 2 and the dissimilar pairs contribute -0.075 (Table S5). Second, this similarity of neighbours is a property of line composition, not of order: adjacent inner tokens share their first glyph, first two glyphs, last glyph, last two or three glyphs, or are identical, exactly as often as tokens of the same line in random order (ratios 0.97 to 1.03; Table S1), whereas Latin and Italian show agreement of final letters (ratios 1.10 and 1.16) and avoid identical neighbours (ratios 0.00 to 0.24). The often-noted immediate repetitions such as *qokeedy qokeedy* (0.82% of adjacent pairs, against 0.16% in German and 0.00% in Latin) occur at the rate implied by random order within lines. Single-glyph edits between adjacent tokens (739 pairs at edit distance 1) are not directional: no edit is significantly more frequent than its inverse (largest $z = 1.8$; Table S2). Finally, a token is no more similar to the token directly above it in the previous line than to a random token of the same page ($P(\text{edit distance} \leq 1)$: 0.044 versus 0.038; minimum over three candidates above, 0.108, versus minimum over three random page tokens, 0.109; Table S3). The neighbour similarity that motivated the copy-and-modify hypothesis is, in these data, entirely page-level locality of the vocabulary (same page 0.038, same language 0.025).

4.3 The text restarts at every line

Table 3 compares the pair formed by the last token of a line and the first token of the next line of the same paragraph with adjacent within-line pairs at the same n . In the controls, cut into lines and paragraphs with the Voynich length distributions, the two are equal (Latin 0.077 versus 0.070, Italian 0.166 versus 0.171, German 0.055 versus 0.060), because the text simply continues. In the Voynich text the line-break pair carries 0.005 bit against 0.062 ± 0.007 for adjacent pairs; with variant glyphs collapsed, 0.022 against 0.145; the same holds within Currier A (0.019 versus 0.052) and B (0.017 versus 0.145). Figure 3 and Table S7 show the same fact from another angle: of all combinations of position classes between consecutive lines of a page, the Latin and Italian controls carry information only in the (last, first) pair (0.065 and 0.178), whereas in the Voynich text that pair is the least informative of all (0.003), below the uniform floor of about 0.013 to 0.022 bit that any two tokens of the same page share. The last token of a line is also the least page-specific token

of the line (I(token; page) 0.055 against 0.113 for middle tokens; Latin at most 0.018 at any position; Table S6). A plain Latin text with one independent sentence per line, which has no continuation across lines by construction, still shows a line-break MI of 0.159 bit (Table 7) because sentence-initial and sentence-final words belong to characteristic classes; the Voynich value is therefore not explained by mere independence of lines in a natural language.

Table 3. Excess mutual information (bits) for the pair (last token of line k , first token of line $k+1$) within paragraphs, compared with adjacent within-line pairs subsampled to the same n (mean $\pm$ s.d. over 8 subsamples). Reference corpora were cut into lines and paragraphs with the Voynich length distributions.

Corpus	n pairs	across line break	adjacent, same n	adjacent inner, same n	across, distance 3	within, distance 3
Voynich (tokens as transcribed)	3,820	0.005	0.062 $\pm$ 0.007	0.071 $\pm$ 0.006	0.025	0.019 $\pm$ 0.005
variant glyphs collapsed	3,820	0.022	0.145 $\pm$ 0.015	0.162 $\pm$ 0.012	0.051	0.051 $\pm$ 0.009
Currier A, collapsed	1,408	0.019	0.052 $\pm$ 0.016	0.058 $\pm$ 0.015	0.013	0.010 $\pm$ 0.013
Currier B, collapsed	2,348	0.017	0.145 $\pm$ 0.020	0.165 $\pm$ 0.010	0.016	0.039 $\pm$ 0.015
Latin	3,488	0.077	0.070 $\pm$ 0.008	0.067 $\pm$ 0.003	0.008	0.007 $\pm$ 0.004
Italian	3,488	0.166	0.171 $\pm$ 0.012	0.166 $\pm$ 0.018	0.000	0.009 $\pm$ 0.004
German	3,079	0.055	0.060 $\pm$ 0.010	0.062 $\pm$ 0.009	0.012	0.001 $\pm$ 0.006

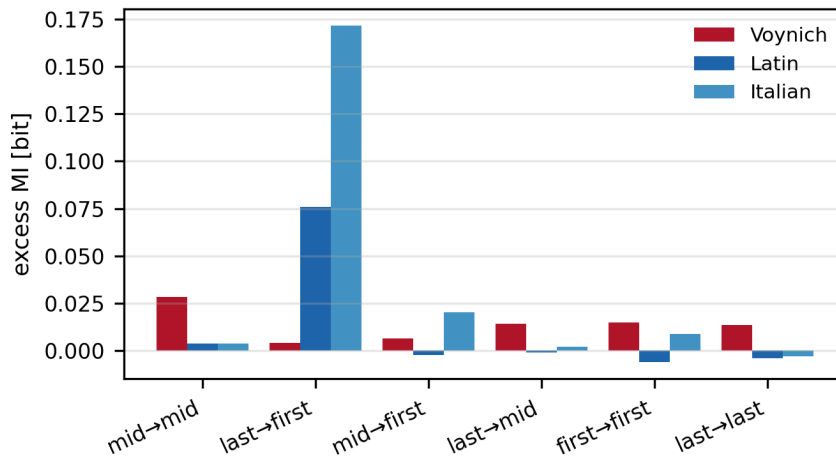


Figure 3. Excess mutual information between tokens of consecutive lines of the same page, by position class. In prose only the (last, first) pair is informative; in the Voynich text it is the least informative pair.

4.4 Token boundaries are predictable from the flanking glyphs

Table 4 and Figure 4 show that one glyph on each side predicts the position of Voynich token boundaries with an information gain of 73.7% ($F_1 = 0.85$), rising to 77.2% with two glyphs. In the four languages the gain from one glyph is 38.7% to 44.9%, and the largest two-glyph value is 61.6% (Latin, whose inflectional endings mark word ends). Latin stripped of its spaces and re-spaced by a fixed glyph rule reproduces the Voynich profile exactly (73.2%, $F_1 = 0.84$). Rozanova and Temerev (2026) report the same predictability with a different estimator (AUC 0.98, 72% reduction of cross-entropy). Together with Section 4.2 this shows that the glyph process continues across spaces with its own transition constraints, and that the spaces fall where those constraints

permit them.

Table 4. Predictability of token boundaries from k glyphs on each side (model fitted on half the lines, evaluated on the other half). Gain = $1 - H(\text{boundary} \mid \text{context})/H(\text{boundary})$. F_1 for predicting a boundary at threshold 0.5.

Corpus	boundary rate	gain, $k = 1$	F_1 , $k = 1$	gain, $k = 2$	F_1 , $k = 2$	gain, $k = 3$
Voynich	0.167	73.7%	0.85	77.2%	0.89	60.2%
Currier A	0.173	–	–	70.4%	0.86	–
Currier B	0.166	–	–	79.5%	0.89	–
Latin	0.147	38.7%	0.55	61.6%	0.79	35.0%
Italian	0.232	43.1%	0.63	53.6%	0.76	19.7%
German	0.189	44.9%	0.66	57.8%	0.77	30.4%
Hebrew	0.206	40.1%	0.57	42.3%	0.65	17.8%
Latin, spaces removed and re-inserted by a glyph rule	0.145	73.2%	0.84	73.3%	0.87	43.2%

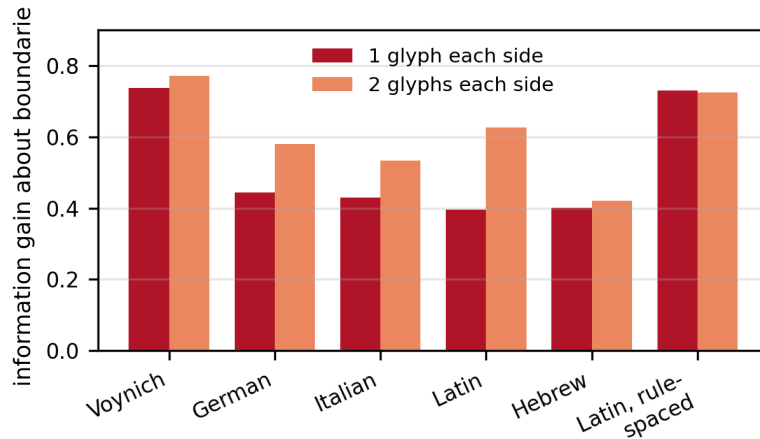


Figure 4. Information gain about token-boundary positions from one or two glyphs on each side, out of sample.

4.5 There is no measurable lexicon

Table 5 gives the lexicon test. Tokens sampled from a glyph trigram model fitted to half of the Voynich tokens are attested in that half as often as real held-out tokens: 0.76 versus 0.79 of tokens and 0.30 versus 0.36 of types (with three glyphs of context, 0.79 versus 0.79). The jackknife interval of the gap includes zero (Table 10). In Latin the corresponding shares are 0.20 versus 0.81 of tokens and 0.03 versus 0.48 of types; in German 0.34 versus 0.83, in Italian 0.52 versus 0.84, in Hebrew 0.36 versus 0.68. Chopping Latin into pieces does not remove the gap: rule-spaced Latin gives 0.28 versus 0.62, syllable pairs 0.09 versus 0.65. The same holds separately for Currier A (0.70 versus 0.71) and B (0.75 versus 0.79). In every language and every chunking of a language the attested strings are a small, specific subset of what the local glyph statistics allow; in the Voynich text a token is, as far as this test can tell, whatever the local glyph statistics produce.

Table 5. Lexicon test. A within-token glyph model with two glyphs of context is fitted on a random half of the tokens; the share of held-out real tokens (types) whose string occurs in the training half is compared with the same share for tokens sampled from the model.

Corpus	real tokens attested	synthetic tokens attested	real types attested	synthetic types attested	bits/glyph real	bits/glyph synthetic
Voynich	0.79	0.76	0.36	0.30	1.90	1.89

Currier A	0.71	0.70	0.34	0.32	1.95	1.93
Currier B	0.79	0.75	0.36	0.31	1.80	1.79
Latin	0.81	0.20	0.48	0.03	2.68	2.67
German	0.83	0.34	0.44	0.07	2.61	2.55
Italian	0.84	0.52	0.44	0.09	2.75	2.70
Hebrew	0.68	0.36	0.40	0.14	3.00	2.81
Latin, rule-spaced	0.62	0.28	0.28	0.09	2.74	2.76
Latin, syllable pairs	0.65	0.09	0.42	0.09	2.99	2.94
Latin, 3-letter chunks	0.96	0.26	0.82	0.25	3.15	3.07

4.6 The entropy profile is that of a low-order automaton

Table 6 and Figure 5 give the conditional entropies of the glyph stream with growing context and their differences to synthetic twins of the same size. Latin, Italian, German and Hebrew all fall below their order-3 twins by -0.30 to -0.19 bit at $h(4)$ (Hebrew -0.13), reflecting the higher-order structure of words and phrases; the Voynich text falls below its order-3 twin by only -0.03, -0.06 and -0.06 bit at $h(3)$, $h(4)$ and $h(5)$. With lzma, which sees all ranges, the Voynich text compresses to 2.40 bit per symbol against 2.58 for its globally fitted order-3 twin, and Latin to 2.58 against 3.11. When the twin is allowed to drift by section, the residual of the Voynich text becomes -0.039 bit per glyph (Table 7), within the noise floor set by the automaton’s own samples (+0.033 and -0.030 for the order-3 and order-4 automata), whereas Latin retains -0.397 and rule-spaced Latin -0.429. Once the local transition rules and their drift are known, no additional redundancy is detectable in the Voynich glyph stream by this measure.

Table 6. Conditional entropy $h(k) = H(x \mid k \text{ previous symbols})$ of the glyph stream (space as a symbol, lines as separate strings), and differences to a synthetic twin sampled from the order-3 model fitted to the same corpus (same number and length of lines). lzma: compressed bits per symbol.

Corpus	$h(1)$	$h(2)$	$h(3)$	$h(4)$	$h(5)$	$h(3) - \text{twin}$	$h(4) - \text{twin}$	$h(5) - \text{twin}$	lzma	lzma twin
Voynich	2.18	1.91	1.82	1.72	1.57	-0.03	-0.06	-0.06	2.40	2.58
Currier A	2.23	1.95	1.83	1.68	1.46	-0.01	-0.03	-0.03	2.62	2.76
Currier B	2.06	1.78	1.70	1.60	1.46	-0.02	-0.04	-0.05	2.33	2.46
Latin	3.38	2.72	2.11	1.66	1.33	+0.00	-0.30	-0.30	2.58	3.11
Italian	3.27	2.77	2.31	1.88	1.43	+0.02	-0.19	-0.18	3.04	3.38
German	3.29	2.63	2.10	1.71	1.38	+0.01	-0.19	-0.17	2.88	3.22
Hebrew	3.63	2.90	2.22	1.60	1.05	+0.09	-0.13	-0.15	3.41	3.83
Latin, rule-spaced	3.21	2.75	2.25	1.83	1.39	+0.01	-0.24	-0.30	2.64	3.21

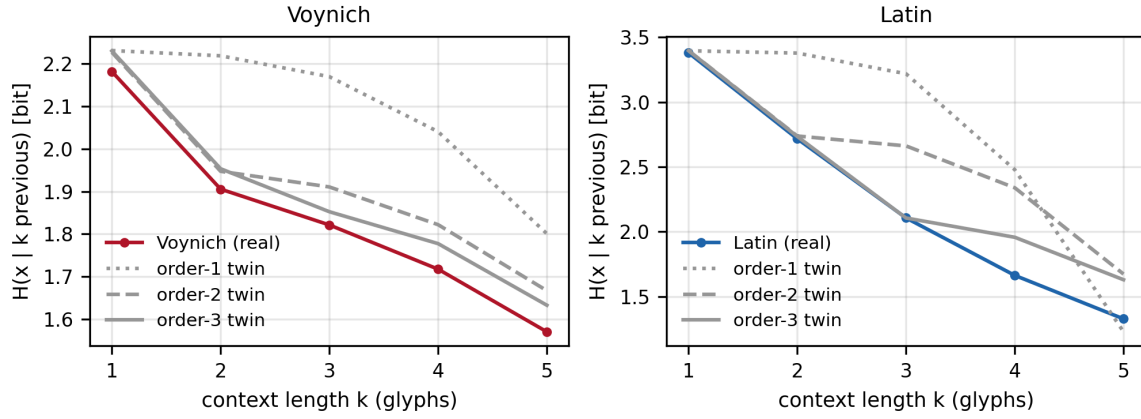


Figure 5. Conditional entropy of the glyph stream given k preceding symbols, for the real text and for synthetic twins sampled from Markov models of order 1, 2 and 3 fitted to the same text with the same line lengths. Left: Voynich; right: Latin cut to the same line shapes.

4.7 Mechanism battery

Table 7 and Figure 6 apply the same battery to the candidate generators of Section 3.7. A glyph automaton of order 3 with section drift reproduces the edge coupling (0.167 against 0.146), the line-break reset, the boundary predictability, the hapax share (0.71 against 0.71), the Zipf slope, the lexicon test, the entropy profile and the residual; its token-level MI (0.050 against 0.078) and its page locality (0.134 against 0.232) fall short. Order 4 closes most of the token-level gap (0.069) at some cost in hapax share (0.64); allowing the parameters to drift at page level ($\lambda = 0.5$, order 4) matches the page locality (0.215) and the token MI (0.076) with edge MI 0.158, residual -0.052 and lexicon shares 0.81 / 0.75; the remaining discrepancy is the hapax share (0.63), which the order-3 variant matches while carrying less token MI (Table S8). No generated line coincides with a real line. We stress that the page-drift automaton is fitted in-sample and is offered only as evidence that page-level parameter variation accounts for the page locality; the out-of-sample evidence is in Section 4.8.

The other mechanisms fail in characteristic ways. Both generators of meaningless text that operate on whole tokens, the table-and-grille and the copy-and-modify generator, have essentially no coupling across the token boundary (0.006 and 0.010 bit, an order of magnitude below the manuscript) because prefix and suffix choices of neighbouring tokens are independent by construction, and both leave large residuals (-0.766 and -1.030) because table reuse and copying create long-range repetition; the copy-and-modify generator also has poor boundary predictability (46.7%) and a lexicon gap (0.57 / 0.23). This agrees with Rozanova and Temerev (2026), who found 0.006–0.009 bit of edge coupling for their implementation of the copy-and-modify model. The verbose cipher with deterministic homophone choice leaks the plaintext’s collocations (token MI 1.237) and has a hapax share of only 0.30. Ciphertext-feedback substitution, the class that can in principle mimic a Markov chain, has too much edge coupling (0.649 and 0.615), unpredictable spaces (36.5%), a lexicon gap (0.39 / 0.27) and residuals of -0.310 and -0.424 because an invertible bounded-memory transformation preserves the plaintext’s redundancy. The plain Latin controls fail the lexicon test and the residual whatever the spacing or line structure; running text and independent sentences also lack the edge coupling (0.037 and 0.035), while rule-spaced Latin reaches an edge MI of 0.134 because the spacing rule itself couples the boundary glyphs, which shows that this one statistic can be produced by a spacing convention alone.

The Naibbe cipher is the strongest competitor. With Latin plaintext and independent sentences per line it matches the entropy profile ($h(2) = 1.90$, $h(4) = 1.70$), the token MI (0.055), the boundary predictability (77.4%), the token length, and, unlike every other plaintext mechanism, the compression residual (-0.065; running text -0.050, Italian -0.065): the entropy of the card draws

hides the redundancy of the plaintext from the compressor, so the residual test is not decisive against homophonic ciphers with high key entropy. The lexicon test separates it only weakly (0.85 / 0.72), because concatenations of table strings look to a glyph trigram model almost like an open inventory. It fails, by a factor of about fifty, the edge coupling (0.003 against 0.146), and it fails the hapax share (0.49 against 0.71; its inventory is closed at 5,978 types), the Zipf slope (-1.00) and the page locality (0.007). Making the choice of table depend on the last glyph of the previous token creates edge coupling, but it is paid for with key entropy: at $w = 1.0$ the edge MI (0.255) exceeds the manuscript's, while the token MI rises to 0.342 and the residual to -0.222 as plaintext structure shows through; at $w = 0.9$ the edge MI is still too small (0.085) while the token MI (0.181) is already too large; at $w = 0.6$, 0.039 and 0.111. No value of w matches both, and the hapax share stays at 0.48. For a cipher, boundary coupling and hidden plaintext redundancy are in competition; for a glyph automaton without plaintext they are not.

Table 7. Mechanism battery. All generators produce text in exactly the line and token shape of the Voynich paragraph text. Edge MI and token MI: excess over within-line permutation; line-break MI: (last, first) pair across line breaks; residual: lzma bits per glyph of the text minus that of an order-3 section-drift twin fitted to the same text (negative = redundancy beyond the automaton). The page-drift automaton is fitted in-sample (see text).

Mechanism	edge MI	token MI	line-break MI	space gain	hapax	Zipf	attested real	attested synth.	h(2)	h(4)	residual	I(token; page)
Voynich (ZL3b)	0.146	0.078	0.005	73.7%	0.71	-0.75	0.79	0.75	1.91	1.72	-0.039	0.232
Glyph automaton, order 3, section drift	0.167	0.050	0.005	73.8%	0.71	-0.77	0.79	0.76	1.92	1.75	+0.033	0.134
Glyph automaton, order 4, section drift	0.149	0.069	-0.001	74.1%	0.64	-0.74	0.81	0.76	1.92	1.69	-0.030	0.138
Glyph automaton, order 4, page drift ($\lambda = 0.5$)	0.158	0.076	0.004	73.8%	0.63	-0.74	0.81	0.75	1.92	1.70	-0.052	0.215
Latin, running text	0.034	0.245	0.076	38.9%	0.59	-0.76	0.80	0.20	2.72	1.65	-0.397	0.062
Latin, rule-spaced	0.134	0.198	0.067	72.8%	0.77	-0.55	0.62	0.29	2.75	1.82	-0.429	0.043
Latin, one independent sentence per line	0.035	0.646	0.159	38.7%	0.35	-0.76	0.86	0.22	2.70	1.57	-0.993	0.050
Table-and-grille generator (after Rugg)	0.006	0.613	0.041	62.8%	0.41	-0.51	0.73	0.55	2.58	1.98	-0.766	0.049
Copy-and-modify generator (after Timm and Schinner)	0.010	0.022	0.042	46.7%	0.62	-0.41	0.57	0.23	2.69	1.97	-1.030	0.912
Verbose cipher, fixed homophone choice	0.409	1.237	0.136	47.4%	0.30	-0.77	0.93	0.64	2.20	1.58	-0.950	0.015
Ciphertext-feedback substitution, 1 glyph/letter	0.649	0.160	0.006	36.5%	0.78	-0.41	0.39	0.27	3.37	1.66	-0.310	0.003

Ciphertext-feedback substitution, 2 glyphs/letter	0.615	0.305	0.009	38.6%	0.70	-0.49	0.54	0.32	2.63	1.45	-0.424	0.007
Naibbe cipher, Latin, independent lines	0.003	0.055	0.002	77.4%	0.49	-1.00	0.85	0.72	1.90	1.70	-0.065	0.007
Naibbe cipher, Latin, running text	0.002	0.078	0.023	77.0%	0.49	-1.01	0.84	0.72	1.91	1.71	-0.050	-0.007
Naibbe cipher, Italian, running text	0.002	0.066	0.029	76.8%	0.47	-1.01	0.86	0.69	1.92	1.73	-0.065	0.001
Naibbe, state-dependent table choice, w = 0.6	0.039	0.111	0.003	76.7%	0.48	-1.02	0.85	0.68	1.93	1.67	-0.101	-0.007
Naibbe, state-dependent, w = 0.9	0.085	0.181	-0.001	76.7%	0.48	-1.01	0.86	0.64	1.92	1.60	-0.148	0.004
Naibbe, state-dependent, w = 1.0	0.255	0.342	0.001	75.4%	0.49	-0.99	0.87	0.66	1.87	1.51	-0.222	0.002

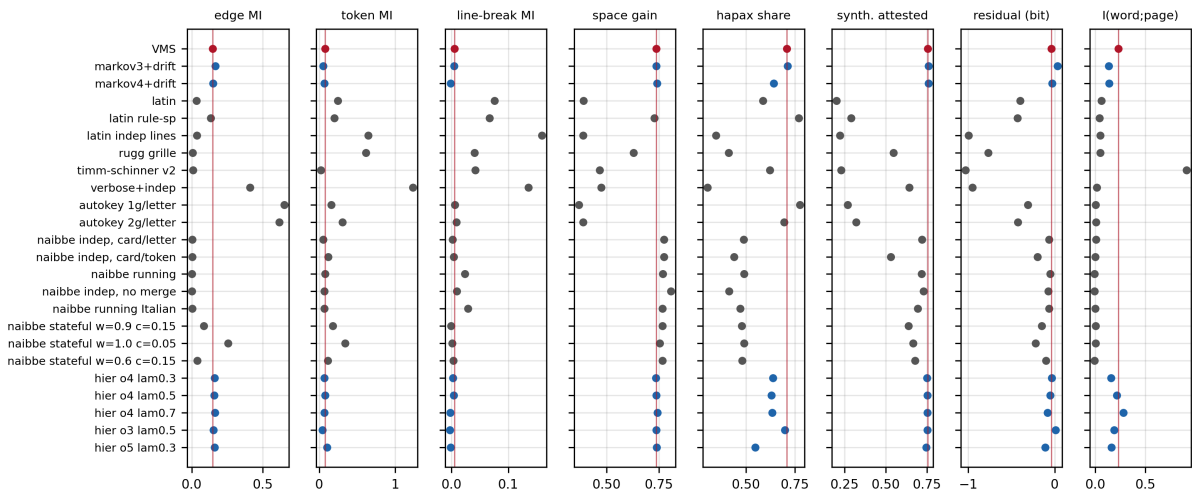


Figure 6. The mechanism battery. Each panel is one statistic; the vertical line marks the Voynich value; blue points are glyph automata, grey points the other generators and controls.

4.8 Robustness, generalization and uncertainty

Table 8 shows that the Takahashi transliteration gives the same profile as ZL3b (edge MI 0.165, line-break MI 0.011, boundary gain 75.2%, hapax 0.68, lexicon 0.83 / 0.79, residual -0.039, page locality 0.245), as does ZL3b with uncertain spaces treated as spaces (edge 0.182, token 0.128, hapax 0.67). Table 9 gives the out-of-sample test: with the automaton fitted per section to the odd-numbered pages and evaluated on the even-numbered pages, the residual is -0.030 (order 3), +0.020 (order 4) and +0.010 for the reversed split, with $h(3)$ residuals of -0.012 to +0.038; the held-out real pages and the twin sampled from the training model agree in edge MI (0.150 versus 0.150), boundary gain (74.1% versus 73.8%), hapax (0.71 versus 0.71), lexicon and entropy profile, and differ in token MI (0.034 versus 0.007) and page locality (0.289 versus 0.168), which is the page-level drift that a section-level model cannot contain. The profile holds within each scribal hand and each Currier language: edge MI 0.079, 0.195 and 0.185 for hands 1, 2 and 3, 0.082 and 0.187 for Currier A and B; line-break MI at zero everywhere; lexicon gap at most 0.04 in any subset.

The edge coupling is weaker in hand 1 and Currier A than in hands 2 and 3 and Currier B, which is a substantive difference between the two languages rather than a failure of the pattern. Table 10 gives the jackknife standard errors of the core statistics. The intervals of the line-break MI and of the lexicon gap include zero, as the null results require; the contrasts with the controls (edge coupling, boundary predictability, lexicon gap) are many standard errors wide. The difference between the edge and the token-level estimate within the manuscript is about two standard errors, but as explained in Section 3.2 the token-level estimate is an attenuated lower bound, and the substantive claim is the location of the information, not the ordering of the two estimates.

Table 8. Robustness to transcription and to the treatment of uncertain spaces.

Transcription	tokens	types	edge MI	token MI	line-break MI	space gain	hapax	attested real / synth.	h(2)	h(4)	residual	I(token; page)
Zandbergen–Landini, uncertain spaces joined (baseline)	32,399	7,832	0.146	0.078	0.005	73.7%	0.71	0.79 / 0.75	1.91	1.72	-0.039	0.232
Zandbergen–Landini, uncertain spaces as spaces	34,859	6,863	0.182	0.128	0.004	74.0%	0.67	0.83 / 0.81	1.89	1.71	-0.044	0.243
Takahashi	34,384	7,054	0.165	0.116	0.011	75.2%	0.68	0.83 / 0.79	1.89	1.71	-0.039	0.245

Table 9. Out-of-sample comparison (automaton fitted per section on the odd-numbered pages, evaluated on the even-numbered pages) and the battery restricted to single scribal hands and to the two Currier languages.

Subset	edge MI	token MI	line-break MI	space gain	hapax	Zipf	attested real / synth.	h(2)	h(4)	I(token; page)
Held-out real pages (even)	0.150	0.034	-0.001	74.1%	0.71	-0.74	0.75 / 0.73	1.89	1.66	0.289
Twin sampled from the odd-page model	0.150	0.007	-0.000	73.8%	0.71	-0.77	0.74 / 0.73	1.92	1.70	0.168
Hand 1 (112 pages)	0.079	0.060	0.001	70.5%	0.71	-0.77	0.71 / 0.71	1.93	1.65	0.129
Hand 2 (46 pages)	0.195	0.087	-0.001	77.4%	0.70	-0.84	0.78 / 0.74	1.72	1.50	0.237
Hand 3 (31 pages)	0.185	0.041	0.006	74.8%	0.72	-0.77	0.71 / 0.69	1.79	1.56	0.182
Currier A	0.082	0.049	0.007	70.6%	0.72	-0.76	0.71 / 0.72	1.95	1.68	0.131
Currier B	0.187	0.081	0.007	76.8%	0.71	-0.78	0.78 / 0.76	1.78	1.60	0.224

Table 10. Leave-one-quire-out jackknife standard errors for the core Voynich statistics (16 quires).

Statistic	estimate	jackknife s.e.	95% interval
edge MI	0.146	0.024	0.099 to 0.193
token MI	0.081	0.024	0.034 to 0.127
line-break MI	0.004	0.010	-0.016 to 0.024
adjacent MI at the line-break n	0.058	0.022	0.015 to 0.101
space gain (k = 1)	0.737	0.018	0.702 to 0.772
hapax share	0.710	0.006	0.698 to 0.722
Zipf slope	-0.753	0.034	-0.819 to -0.687
attested, real	0.792	0.026	0.742 to 0.843
attested, synthetic	0.754	0.025	0.704 to 0.803
lexicon gap (real – synthetic)	0.039	0.021	-0.003 to 0.080

h(2)	1.905	0.047	1.812 to 1.998
h(4)	1.717	0.025	1.669 to 1.765

4.9 Drift and scribal hand

Table 11 and Figure 7 show that the glyph-bigram distributions of two pages of the same section and Currier language diverge more the farther apart they are in the binding: for Currier A, from 0.124 at one or two pages to 0.161 beyond forty pages (0.193 across sections); for B, from 0.126 to 0.147 (0.169). The section-drift twin is flat at 0.093 to 0.094 within a section, and Latin in book order is nearly flat (0.183 to 0.192). The Voynich text therefore has both a page-specific component (its nearest pages are already more different than the twin's) and a slow drift with folio distance. Table 12 relates the variation to the scribal hand. Within Currier B, pages of the same section and hand differ by 0.135, of the same section and different hands by 0.153, of different sections and the same hand by 0.172; but sections occupy contiguous folio ranges, so the section effect is confounded with drift. Within Herbal-B, the only section written by several hands (32 pages; hands {'2': 20, '5': 6, '3': 6}), pairs at a folio distance of at most six pages differ by 0.138 when written by the same hand and by 0.182 when written by different hands (48 and 22 pairs); at distances beyond twenty pages the values are 0.157 and 0.165. The bigrams that distinguish hand 2 from hand 3 within this one section are the same that distinguish Currier B from A (hand 2 over-uses *y_* (+1.21), *ke* (+0.92), *he* (+0.88), *dy* (+0.79), *ed* (+0.68), *eo* (+0.65); hand 3 over-uses *in* (+1.19), *n_* (+1.17), *ai* (+0.94), *ot* (+0.87), *ta* (+0.79), *ka* (+0.74); differences in percentage points of all bigrams, an underscore marking the space). That hands and statistical languages coincide has been known since Currier (1976) and was confirmed palaeographically by Davis (2020); what is new here is the controlled comparison: at fixed section, language and folio distance, the hand effect exceeds the section effect. The observed variation therefore follows the scribal hand more than the illustration section, which makes scribal practice a plausible source of the parameter drift. This does not exclude a cipher, since a scribe using the same cipher could have personal preferences in table choice, abbreviation or error; it does mean that the drift need not be explained by content.

Table 11. Mean Jensen–Shannon divergence between the glyph-bigram distributions of two pages (450-glyph samples) as a function of their folio distance, within the same section and Currier language.

Pages	d = 1–2	3–6	7–14	15–40	> 40	other section
Voynich, Currier A	0.124	0.145	0.150	0.150	0.161	0.193
Voynich, Currier B	0.126	0.137	0.134	0.147	0.147	0.169
Section-drift twin, A	0.093	0.093	0.094	0.093	0.094	0.148
Section-drift twin, B	0.086	0.085	0.085	0.092	0.092	0.122
Latin in book order (chunked like A)	0.183	0.188	0.195	0.192	0.192	0.193

Table 12. Mean Jensen–Shannon divergence between pages grouped by scribal hand (palaeographic assignment of Davis 2020 as recorded in the ZL metadata) and illustration section.

Pairs	same section, same hand	same section, different hand	different section, same hand	different section, different hand
Currier B, all sections	0.135	0.153	0.172	0.168
Herbal-B, folio distance ≤ 6 pages	0.138	0.182	–	–
Herbal-B, folio distance > 20 pages	0.157	0.165	–	–

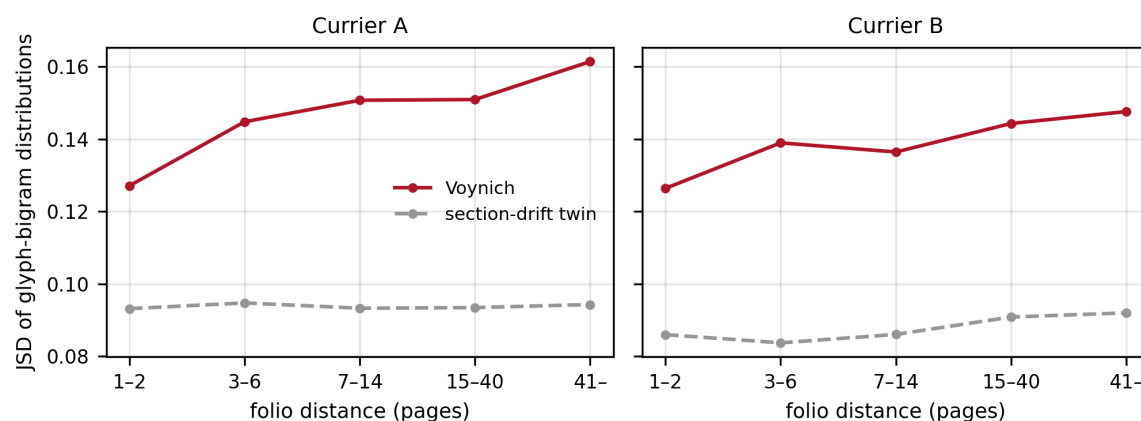


Figure 7. Jensen–Shannon divergence between the glyph-bigram distributions of two pages of the same section as a function of their folio distance, for the manuscript and for the section-drift twin.

4.10 Labels

Table 13 shows that label tokens recur across the pages of a section less often than randomly drawn paragraph-text tokens of the same pages: in the pharmaceutical section 0.090 against 0.364 (219 types among 234 tokens), in the astronomical section 0.028 against 0.273, in the biological section 0.115 against 0.342, in the zodiac section 0.276 against 0.363 (271 types among 330 tokens on twelve sign pages). If the zodiac labels were numerals of days or degrees they would recur on every sign page; only 36 of 271 label types occur on two or more of the twelve sign pages, and none on more than 4. Labels behave as unique identifiers, unlike numerals and unlike text tokens, and, in the pharmaceutical and astronomical sections, avoid even the frequent token types. Whether they are names or the output of the same process with repetition avoided cannot be decided from these data; but they are not what the paragraph-text process produces when sampled at random.

Table 13. Recurrence of label tokens across the pages of a section, compared with equally many randomly drawn paragraph-text tokens from the same pages (mean of 20 draws).

Section	pages	label tokens	label types	share recurring on another page	null: text tokens of the same pages
Zodiac	12	330	271	0.276	0.363
Astronomical	5	142	136	0.028	0.273
Pharmaceutical	15	234	219	0.090	0.364
Biological	11	122	111	0.115	0.342
Cosmological	5	196	181	0.020	0.277

4.11 Does the text carry information about the illustrations?

This is the test that separates “the text needs no words to be modelled” from “the text carries no measurable information about the page it is on”. Table 14 and Figure 8 give Mantel correlations between text distance and illustration-feature distance across herbal pages of one hand. For the 85 Herbal-A pages of hand 1 (3,237 pairs at three or more folios apart) the correlation is $r = -0.055$ ($p = 0.79$, 10,000 permutations) at the token level and $r = -0.070$ ($p = 0.87$) at the glyph-bigram level; no single image feature correlates (all $p \geq 0.22$, Table S10). The same token-level distance does detect the folio drift ($r = +0.115$ with folio distance, $p = 0.001$), so the null result is not a failure of the text representation. For the 25 Herbal-B pages an unstratified test gives $r = +0.223$ ($p = 0.046$), but the effect disappears when the permutation is stratified within hand ($p = 0.076$), when only same-hand pairs are used ($r = +0.144$, $p = 0.14$) or when hand 2 alone is considered (20 pages, $r = +0.143$, $p = 0.14$): it was carried by the hand, to which both painting style and

glyph statistics are tied through the quire.

The sensitivity of the test is calibrated in Table 15. Adding to each Herbal-A page k pseudo-tokens that are deterministic functions of its image features, the test becomes significant at $k = 10$ ($r = +0.195$, $p = 0.001$), strong at $k = 20$ ($r = +0.366$), and is not significant at $k = 5$ ($r = +0.053$, $p = 0.21$); the mean page has 77 tokens, so ten tokens are about 11% of a page. The null result therefore excludes descriptions in which about a tenth or more of the tokens of a page are determined by the measured colour features of its drawing; it says nothing about content that is not visible in the drawing (names, uses, properties) or about descriptions below that share.

Table 14. Mantel tests between text distance and illustration-feature distance across herbal pages (pairs closer than 3 folios excluded).

Page set	text representation	Mantel r	p (permutation)	text vs folio distance r (p)
Herbal-A, hand 1 (85 pages)	token frequencies	-0.055	0.79	+0.115 (0.001)
	glyph-bigram frequencies	-0.070	0.87	–
Herbal-B, hands 2 and 5 (25 pages), unstratified	token frequencies	+0.223	0.046	+0.163 (0.044)
permutation within hand	token frequencies	+0.223	0.076	–
same-hand pairs only	token frequencies	+0.144	0.14	–
hand 2 only (20 pages)	token frequencies	+0.143	0.14	–
hand 2 only	glyph-bigram frequencies	-0.036	0.58	–

Table 15. Sensitivity control: image-determined pseudo-tokens added to each Herbal-A page (mean page length 77 tokens).

tokens added per page	Mantel r	p
2	-0.036	0.711
5	+0.053	0.209
10	+0.195	0.001
20	+0.366	0.001

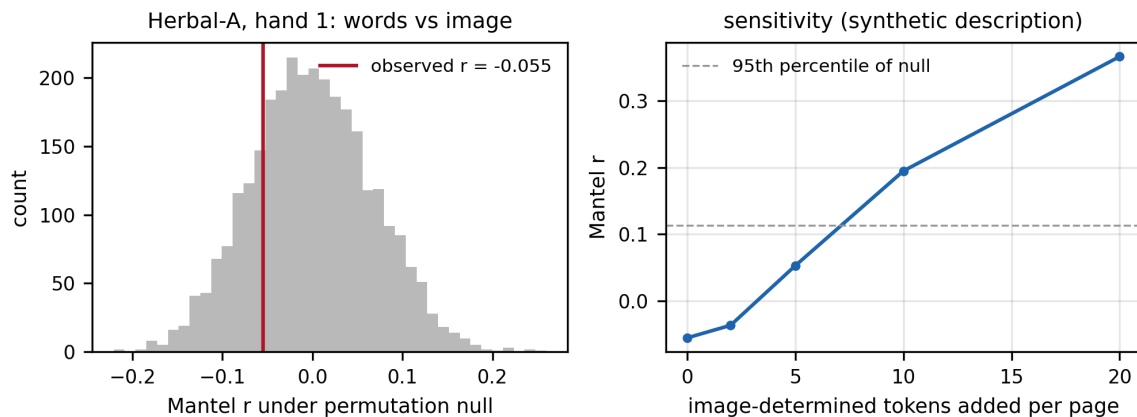


Figure 8. Left: Mantel correlation between token-frequency distance and illustration-feature distance across Herbal-A pages of hand 1, against its permutation null. Right: sensitivity of the same test when image-determined pseudo-tokens are added to each page.

Table 16 gives the secondary section-prediction analysis. Within hand 1 (herbal versus

pharmaceutical), tokens (0.79 to 0.81) and glyph bigrams (0.75 to 0.80) both stay below the majority rate of 0.86. Within hand 2 (biological versus herbal), glyph bigrams predict the section (0.95 without and 0.79 with a fifteen-folio exclusion radius) far better than tokens (0.74 and 0.42), and the nearest remaining folio alone predicts it at 1.00 and 0.49. Within hand 3 tokens stay at the majority rate. Where the text carries any section signal at all, it lies in the parameters of the local glyph process rather than in token identities beyond them, and it cannot be separated from folio drift with this design.

Table 16. Leave-one-page-out nearest-centroid prediction of the illustration section within a scribal hand, from token frequencies and from glyph-bigram frequencies, excluding pages within the given folio radius; compared with the majority rate and with predicting from the nearest remaining folio.

Hand (sections)	majority rate	excluded radius (folios)	tokens	glyph bigrams	nearest folio only	n
Hand 1 (herbal vs pharmaceutical)	0.86	0	0.79	0.75	0.97	111
	0.86	5	0.79	0.75	0.84	111
	0.86	15	0.81	0.80	0.85	86
Hand 2 (biological vs herbal)	0.51	0	0.74	0.95	1.00	39
	0.51	5	0.68	0.89	0.95	37
	0.51	15	0.42	0.79	0.49	19
Hand 3 (stars vs herbal)	0.77	0	0.81	0.77	0.97	31
	0.77	5	0.77	0.77	0.71	31
	0.77	15	0.77	0.65	0.03	31

5. Discussion

5.1 What is established

Four properties of the Voynich text are established by tests with marginal-preserving null models, identical sample structure for the controls, and jackknife uncertainty: ordering information is concentrated at token boundaries; the text restarts at each line; token boundaries are predictable from the flanking glyphs to the degree of rule-inserted spaces; and there is no measurable lexicon. These hold on two transliterations, with either treatment of uncertain spaces, within each scribal hand and each Currier language. A fifth property is positive: a glyph automaton of order 3–4 with parameters drifting by section or page, restarted at each line, reproduces every statistic of the battery, generalizes to held-out folios, and leaves no compression residual beyond its own noise. The evidence and the interpretation should be kept apart. The evidence is that all tested observable structure is compatible with such a local process. The interpretation that the text was produced by such a process is not established: the automaton is a positive null model, and another process with the same statistical footprint is not excluded. What the results do exclude is any mechanism in the tested classes whose footprint differs.

5.2 Three possible end states

It is useful to distinguish three end states for research on this text. (a) Decipherment: a transformation that yields meaningful plaintext on unseen folios and explains labels or illustrations. Nothing here points to one, and Sections 4.5 and 4.11 constrain it: it would have to map an open, lexicon-free token inventory onto a lexicon, and it would have to recover page content that leaves no trace in the colour composition of the drawings. (b) An identified generator: a concrete, historically plausible procedure that predicts unseen text as well as the manuscript predicts itself and reproduces its peculiarities. The automaton does this statistically but is a null model, not a

historical procedure; the published procedures (table and grille, copy and modify) do not. (c) No demonstrable semantics: after controlling for hand, section and folio, nothing external can be predicted from the text, and a local automaton reaches, on held-out folios, essentially all the predictive performance that is attainable. The present results are of type (c), with a quantified sensitivity for the external test. Type (c) is not a proof of meaninglessness; it is a statement about what is measurable.

5.3 What a cipher would need

The pattern of failures in Table 7 delineates the requirements. A cipher of a natural-language plaintext would need (i) state-dependent choice of glyph strings, to produce the 0.146 bit of coupling across token boundaries; (ii) high key entropy, so that the plaintext’s redundancy and lexicon disappear from the ciphertext, as the Naibbe cipher achieves with its card draws; (iii) an open inventory of thousands of strings with a hapax share of 0.71 at 32,399 tokens; (iv) no continuation across line breaks, which the plaintext must supply, since even a Latin text of independent sentences leaves 0.159 bit at the line break; and (v) parameters that drift along the folio sequence and with the scribe. Requirements (i) and (ii) pull in opposite directions in the state-dependent Naibbe variants: coupling bought with state costs key entropy, which exposes plaintext structure in the token MI and in the residual. A procedure satisfying all five would be, by construction, a drifting glyph automaton whose transitions are partly driven by the plaintext, that is, a mimic function in the sense of Wayner (1992). We know of no published cipher with these properties; whether one would be historically plausible is a different question from whether it is statistically possible, and our tests answer only the second.

5.4 Relation to previous work

Our Sections 4.2 to 4.4 agree quantitatively with Rozanova and Temerev (2026), who used the same transliteration and different estimators, and with the older observations of Currier (1976) on the line as a unit and of Lindemann and Bown (2020) on the unusually strong local constraint of the glyphs. The lexicon test, the residual against a drifting twin, the out-of-sample generalization, the mechanism battery with its positive null model, the elimination of ciphertext-feedback substitution and of the state-dependent Naibbe variants, the drift with folio distance, the controlled hand-versus-section comparison, the label recurrence test and the external information test are, to our knowledge, new. Our results are consistent with Greshko (2025) and Kinnison (2026) in that entropy profiles and token-level statistics are reproducible by ciphers, and add that boundary coupling, hapax richness and page locality are not reproduced by the Naibbe cipher, so that the three constraints have to be met together. They are consistent with Timm and Schinner (2020) in the sense that a simple local process suffices, but not with their specific mechanism: the copy-and-modify model has no boundary coupling and no vertical copy signature is detectable once page-level vocabulary locality is controlled. The long-range structure reported by Montemurro and Zanette (2013) corresponds, in our measurements, to the page- and section-level drift of the glyph statistics, which a drifting automaton reproduces and which follows the scribal hand more than the illustration section.

5.5 Objections we anticipate

- *Transcription artefacts.* Spaces and glyph identities are transcriber judgements. The battery gives the same profile on the Takahashi transliteration and with uncertain spaces treated either way (Table 8), and the edge coupling is, if anything, stronger with uncertain spaces treated as spaces.
- *Estimator bias.* Plug-in MI is biased and the bias depends on sparsity. All comparisons are excess values over marginal-preserving nulls at equal n , and the qualitative contrasts (inverted hierarchy, line reset, lexicon) are large relative to the jackknife errors. The attenuated token-level values are not interpreted as absolute quantities.

- *In-sample fitting.* The page-drift automaton is in-sample. The claims of generalization rest on the odd/even split (Table 9), where the training model was never shown the evaluated pages; the only statistics it misses are those that depend on page-level parameters.
- *Section versus drift.* Sections occupy contiguous folio ranges, so section effects cannot be separated from drift in this manuscript. We report the confound rather than resolve it, and use the one section with several hands to show that the hand effect is larger at fixed distance.
- *The generators are approximations.* Our table-and-grille and copy-and-modify implementations follow the published descriptions but are not the authors' code; their structural failure (independent choice of neighbouring boundary glyphs) follows from the constructions, not from parameters. The Naibbe implementation uses the published tables and documented procedure but was not validated against the author's outputs.
- *Crude image features.* Colour shares are a weak proxy for the content of a herbal illustration. The sensitivity control quantifies what the test could have seen; better annotations (plant parts, flower shapes, number of plants, zodiac signs) would raise the sensitivity, and the test code accepts any per-folio feature vector.
- *Unrepresentative controls.* Classical Latin prose, Italian and German verse, and consonantal Hebrew are not a medieval herbal in abbreviated Latin. The inverted MI hierarchy and the lexicon gap hold for all four and for every chunking of Latin we tried, but a control from a heavily abbreviated medieval register remains desirable.

6. Limitations

The transliterations, not the manuscript, are the object of measurement. Only folios 1–58 were used for the illustration test, and the colour features capture leaf, flower and root areas, not plant identity or use. The folio order is the binding order. The residual measure uses a general-purpose compressor and is meaningful only relative to a twin of the same text. The automaton reproduces the battery but is not a historical procedure, and its page-level variant is fitted in-sample. Our Naibbe implementation was not verified against the original outputs, and the copy-and-modify and table-and-grille generators are our reconstructions. The tests have finite sensitivity, quantified for the external test (about ten image-determined tokens per page) and only implicitly, through the jackknife, for the others. None of these limitations bears on the direction of the main contrasts, but each bounds how far the conclusions reach.

7. Conclusion

Under a common battery of information-theoretic tests with marginal-preserving null models and identically structured controls, the Voynich text shows ordering information at token boundaries rather than between tokens, a restart at every line, boundary positions predictable from the flanking glyphs, and no measurable lexicon. A glyph automaton of low order, with parameters that drift by section, page and scribal hand and that restarts at every line, reproduces all of these statistics, generalizes to held-out folios, and leaves no compression residual beyond its own noise; published generators of meaningless text and several substitution ciphers, including the Naibbe cipher and a state-dependent extension of it, each fail at least two statistics. Within a single hand and section the text carries no measurable information about the colour composition of the illustrations, with a sensitivity of about a tenth of the tokens of a page. All tested observable structure is therefore compatible with a local, low-order glyph generator, and neither lexical structure nor image-related semantics is measurable beyond it. This does not show that the text is meaningless; it shows what any proposed reading or generating procedure must now account for, and it provides a null model, code and logs against which such proposals can be tested.

Data and code availability

The parser, the test battery, all generators including the Naibbe reimplementations, every analysis script, the logs from which every number in this paper is extracted, the extracted numbers, the figure scripts and the manuscript source are provided in the accompanying repository [URL to be added on submission]. The transliterations are those of voynich.nu (Zandbergen and Landini 2025; Takahashi 1999); the Naibbe tables are those published with Greshko (2025) under the author’s licence, which requires attribution; the reference texts are from Project Gutenberg, The Latin Library and the Sefaria API; the facsimile is the Beinecke Library’s digital reproduction of MS 408.

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